

Lecture 8

Orthonormal Bases, Gram–Schmidt, and Riesz

The core uses finite-dimensional inner-product spaces over $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ in the physics convention (conjugate-linear in the first slot). Tags mark the flavour; a ★ marks a harder one.

Core tutorial route

Work **8.2, 8.3, 8.5, 8.17, 8.8, 8.9, and 8.19** in that order (about 80–100 minutes before discussion). This seven-problem route covers coordinates and Parseval, real and complex Gram–Schmidt, an orthogonal complement/dimension check, best approximation, least-squares transfer, and a genuine finite Riesz explanation with the complex bra–ket convention.

Additional practice: 8.1, 8.4, 8.6–8.7, 8.10–8.14, 8.16, 8.18, 8.20, 8.24. **Bridge:** 8.15, 8.21–8.22 (integral/Frobenius models and a locally defined adjoint preview). **Extension:** 8.23 (codimension-one forward link). Extension tasks do not count toward core progress.

Orthonormal bases and Gram–Schmidt

Exercise 8.1 (computational) Apply Gram–Schmidt to $v_1 = (1, 1, 1)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 0, 0)$ in $\mathbb{R}^3$ with the dot product.

Answer. $e_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$, $e_2 = \frac{1}{\sqrt{6}}(1, 1, -2)$, $e_3 = \frac{1}{\sqrt{2}}(1, -1, 0)$.

Exercise 8.2 (computational) Expand $v = (2, 3) \in \mathbb{R}^2$ in the orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1, 1)$, $e_2 = \frac{1}{\sqrt{2}}(1, -1)$, and verify Parseval's relation.

Answer. $v = \frac{5}{\sqrt{2}}e_1 - \frac{1}{\sqrt{2}}e_2$; Parseval: $\frac{25}{2} + \frac{1}{2} = 13 = \|v\|^2$.

Exercise 8.3 (computational) Apply Gram–Schmidt to $v_1 = (3, 4)$, $v_2 = (1, 0)$ in $\mathbb{R}^2$ with the dot product.

Answer. $e_1 = \frac{1}{5}(3, 4)$, $e_2 = \frac{1}{5}(4, -3)$.

Exercise 8.4 (computational) Expand $v = (3, 0, 0) \in \mathbb{R}^3$ in the orthonormal basis $e_1 = \frac{1}{3}(1, 2, 2)$, $e_2 = \frac{1}{3}(2, 1, -2)$, $e_3 = \frac{1}{3}(2, -2, 1)$, and check Parseval's relation.

Answer. $v = e_1 + 2e_2 + 2e_3$; Parseval: $1 + 4 + 4 = 9 = \|v\|^2$.

Exercise 8.5 (computational) Orthonormalize $v_1 = (1, i)$, $v_2 = (0, 1)$ in $\mathbb{C}^2$ (physics convention, conjugate-linear in the first slot).

Answer. $e_1 = \frac{1}{\sqrt{2}}(1, i)$, $e_2 = \frac{1}{\sqrt{2}}(i, 1)$.

Exercise 8.6 (proof) Prove that an orthonormal list $e_1, \dots, e_m$ is linearly independent directly from the definition.

Exercise 8.7 (proof) Let $e_1, \dots, e_n$ be an orthonormal basis of V . Prove that for all $u, v \in V$,

$$\langle u, v \rangle = \sum_{k=1}^n \overline{\langle e_k, u \rangle} \langle e_k, v \rangle,$$

so the coordinate map $v \mapsto (\langle e_1, v \rangle, \dots, \langle e_n, v \rangle)$ preserves inner products.

Projections and best approximation

Exercise 8.8 (computational) Find the point of the line $U = \text{span}((1, 2, 2)) \subset \mathbb{R}^3$ closest to $b = (3, 0, 0)$.

Answer. $P_U b = \frac{1}{3}(1, 2, 2)$.

Exercise 8.9 (computational) Fit a line $y = c_0 + c_1 x$ to the data $(1, 1), (2, 2), (3, 2)$ by least squares.

Answer. $y = \frac{2}{3} + \frac{1}{2}x$.

Exercise 8.10 (computational) Find the orthogonal projection of $b = (2, 4) \in \mathbb{R}^2$ onto the line $U = \text{span}((1, 1))$, and the distance from b to U .

Answer. $P_U b = (3, 3)$; distance $\|b - P_U b\| = \sqrt{2}$.

Exercise 8.11 (computational) Project $b = (3, 0, 0) \in \mathbb{R}^3$ onto the plane U with orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$.

Answer. $P_U b = (2, 1, 1)$; distance $\sqrt{3}$.

Exercise 8.12 (computational) Solve the inconsistent system $Ax = b$ in the least-squares sense, where $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $b = (1, 1, 0)$.

Answer. $\hat{x} = (\frac{1}{3}, \frac{1}{3})$.

Exercise 8.13 (proof) Suppose $P = P_U$ is an orthogonal projection. Prove $P^2 = P$ and that $v - Pv \perp Pv$ for every v .

Exercise 8.14 (proof) Let P_U be the orthogonal projection onto a subspace U with orthonormal basis $e_1, \dots, e_m$. Prove that $\langle P_U v, w \rangle = \langle v, P_U w \rangle$ for all $v, w \in V$.

Exercise 8.15 (★) On $\mathcal{P}_2$ with $\langle p, q \rangle = \int_{-1}^1 \overline{p} q dx$, find the polynomial of degree ≤ 1 closest to $f(x) = x^2$, and verify that the error is orthogonal to $\mathcal{P}_1$.

Hint. Project x^2 onto $\mathcal{P}_1 = \text{span}(1, x)$ using the orthonormal $p_0 = \frac{1}{\sqrt{2}}$, $p_1 = \sqrt{\frac{3}{2}}x$ from the lecture; an odd integrand over $[-1, 1]$ integrates to 0.

Answer. The closest polynomial is the constant $\frac{1}{3}$; the error is $x^2 - \frac{1}{3}$.

Riesz and orthogonal complements

Exercise 8.16 (computational) Find the vector $w \in \mathbb{C}^2$ representing the functional $\varphi(v) = 2v_1 - iv_2$ in the form $\varphi(v) = \langle w, v \rangle$.

Answer. $w = (2, i)$.

Exercise 8.17 (computational) For $U = \text{span}((1, 0, 1)) \subset \mathbb{R}^3$, describe $U^\perp$ and verify $\dim U + \dim U^\perp = 3$.

Answer. $U^\perp = \text{span}((1, 0, -1), (0, 1, 0))$, $\dim U^\perp = 2$; $1 + 2 = 3$.

Exercise 8.18 (computational) Find $w \in \mathbb{R}^3$ representing the functional $\varphi(v) = 2v_1 - v_2 + 3v_3$ in the form $\varphi(v) = \langle w, v \rangle$.

Answer. $w = (2, -1, 3)$.

Exercise 8.19 (proof) State and prove the finite-dimensional Riesz representation theorem for a linear functional $\varphi: V \rightarrow \mathbb{F}$, including its hypotheses, existence, and uniqueness. Then explain why $w \mapsto \langle w, \cdot \rangle$ is conjugate-linear over $\mathbb{C}$ but linear over $\mathbb{R}$.

Answer. For finite-dimensional inner-product V over $\mathbb{F}$, every linear $\varphi: V \rightarrow \mathbb{F}$ is uniquely $\langle w, \cdot \rangle$; the representing-vector-to-functional map conjugates complex scalars.

Exercise 8.20 (computational) For $U = \text{span}((1, 0, 1, 0), (0, 1, 0, 1)) \subset \mathbb{R}^4$, give a basis of $U^\perp$ and check $\dim U + \dim U^\perp = 4$.

Answer. $U^\perp = \text{span}((1, 0, -1, 0), (0, 1, 0, -1))$, $\dim U^\perp = 2$; $2 + 2 = 4$.

Exercise 8.21 (computational) On $\mathcal{M}_2(\mathbb{R})$ with the Frobenius product $\langle A, B \rangle = \text{tr}(A^\dagger B)$, find the matrix w representing $\varphi(A) = 2A_{11} - A_{12} + A_{22}$ as $\varphi(A) = \langle w, A \rangle$.

Answer. $w = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}$.

Exercise 8.22 (★) **Forward bridge.** For a matrix/operator T in an ONB, define its adjoint $T^\dagger$ by $\langle T^\dagger u, v \rangle = \langle u, Tv \rangle$ (equivalently, conjugate transpose of the matrix). Prove that $\text{tr}(T^\dagger T) = \sum_{j,k} |\langle e_j, Te_k \rangle|^2 \geq 0$, with equality iff $T = 0$.

Hint. Write the matrix entries of T as $\langle e_j, Te_k \rangle$ and recognise $\text{tr}(T^\dagger T)$ as the sum of their squared moduli.

Exercise 8.23 (★) Suppose V is finite-dimensional and φ is a linear functional on V represented by w via $\varphi(v) = \langle w, v \rangle$. Prove that $\text{null } \varphi = (\text{span}(w))^\perp$ and, if $w \neq 0$, that $\dim \text{null } \varphi = \dim V - 1$.

Hint. $\varphi(v) = 0$ is exactly $\langle w, v \rangle = 0$. For the dimension, apply the finite-dimensional orthogonal-complement formula to $\text{span}(w)$.

Exercise 8.24 (proof) Let U be a subspace of an inner product space V . Prove directly from the definitions that $U \subseteq (U^\perp)^\perp$.